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Education (CBE)**

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GRADE 12

SENIOR SCHOOL

CHEMISTRY

BOOK 1

**CONCEPTS, APPLICATION &
EXAMINATION PREPARATION**



EXPERIMENT
ANALYSE
APPLY
EXCEL



Clear
Explanations



Practical
Applications



Worked
Examples



Investigations



Topic &
Mixed Tests



5 National CBE
Model Papers



Marking Schemes
& Quick Reference

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GRADE 12

SENIOR SCHOOL CHEMISTRY

BOOK 1

CONCEPTS, APPLICATION & EXAMINATION PREPARATION

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Learner-Friendly • Application-Based • Examination-Focused

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Peak Achievers Online Academy

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Disclaimer: This is an independently developed supplementary learning and examination-preparation resource. It is not an official publication of KICD, KNEC or the Ministry of Education.

Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.



To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

— Hezekiah Omweri Nyatuka

Preface

Chemistry becomes easier when a learner understands why processes occur rather than merely memorising facts and equations. Peak Achievers Grade 12 Senior School Chemistry Book 1 is designed as a concise, clear and application-oriented resource for Kenya's Competency-Based Education context.

The book emphasises chemical concepts in real situations: industrial processes, investigations, calculations, environmental issues, interpretation and application. Notes are intentionally brief but specific, supported by equations, diagrams, worked examples and learner-support features.

Assessment tasks encourage learners to investigate, interpret, calculate, explain, predict, justify, design, evaluate and propose.

UNDERSTAND → APPLY → PRACTISE → ACHIEVE

How to Use This Book

REMEMBER - key fact | HINT - approach | CAUTION - common error | SPOTLIGHT - real-life chemistry | INVESTIGATION - practical inquiry | THINK BIG - higher-order CBE | EXAM FOCUS - response technique

Do not simply read Chemistry. Work with it. Study each short section, understand the concept, follow the equations and diagrams, connect the chemistry to its applications, then attempt the revision questions before checking the marking guidance.

Calculations: FORMULA → SUBSTITUTION → WORKING → ANSWER + UNIT

Explanations: CLAIM + CHEMICAL EVIDENCE + REASONING

FREE SAMPLE - NOT FOR SALE

CONTENTS

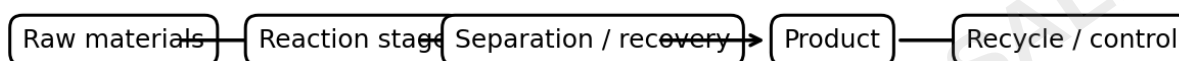
1.0 INORGANIC CHEMISTRY

- 1.1 Industrial Manufacture of Inorganic Chemicals

EXAM FOCUS: When explaining an industrial condition, use: CONDITION -> CHEMICAL REASON -> INDUSTRIAL ADVANTAGE.

REMEMBER: Industrial conditions are chosen as compromises between rate, yield, energy use, safety and cost.

Industrial Manufacture - Process Thinking



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

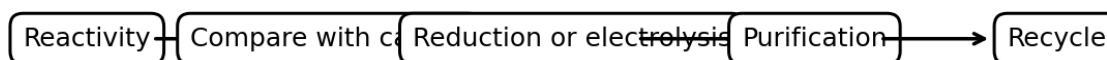
- 1.2 Carbon and Extraction of Metals

THINK BIG: Compare extraction from ore with recycling in terms of energy, resources, waste and environmental disturbance.

CAUTION: Limestone does not reduce iron ore. It provides CaO, which removes silica impurity as slag.

HINT: For extraction questions, first locate the metal relative to carbon in the reactivity series.

Choosing a Metal Extraction Method



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

2.0 PHYSICAL CHEMISTRY

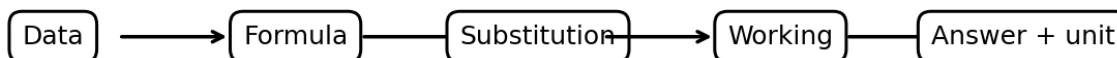
- 2.1 Energy Changes in Physical and Chemical Processes

EXAM FOCUS: Calculations: FORMULA -> SUBSTITUTION -> WORKING -> ANSWER + UNIT.

CAUTION: Keep the sign of ΔH consistent: exothermic $\Delta H < 0$; endothermic $\Delta H > 0$.

REMEMBER: $q = mc\Delta T$. For an exothermic reaction, heat gained by the solution corresponds to heat released by the reaction.

Peak Achievers Calculation Route



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

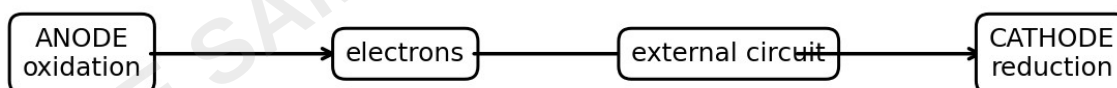
2.2 Electrochemistry

SPOTLIGHT: Battery chemistry should be linked to raw-material extraction, use, collection and safe recycling/disposal.

CAUTION: Do not multiply standard electrode potentials when multiplying half-equations.

REMEMBER: OIL RIG: Oxidation Is Loss; Reduction Is Gain of electrons.

Electrochemical Cell



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

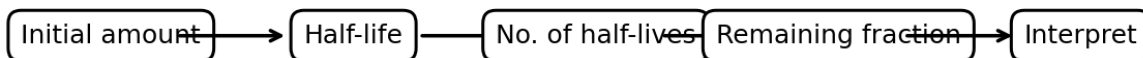
2.3 Nuclear Chemistry

THINK BIG: A useful medical radioisotope must remain active long enough for the procedure without remaining radioactive unnecessarily long.

CAUTION: Ordinary changes in temperature do not significantly change radioactive half-life.

HINT: When numbers are simple, count the number of halvings before using the half-life formula.

Radioactivity Problem Route



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

3.0 ORGANIC CHEMISTRY

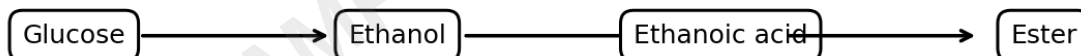
- 3.1 Alkanols

SPOTLIGHT: Compare fermentation and ethene hydration using source of raw material, rate, purity and sustainability.

INVESTIGATION: Relate fermentation conditions to evidence: glucose, yeast, suitable warmth and limited oxygen.

REMEMBER: Alkanols contain the hydroxyl functional group, -OH.

Key Alkanol Transformations



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

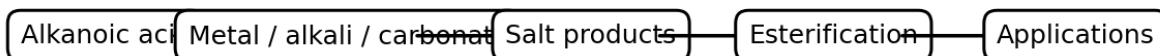
- 3.2 Alkanoic Acids

EXAM FOCUS: For practical questions, write OBSERVATION -> INTERPRETATION -> CONCLUSION.

CAUTION: Do not confuse acid strength with concentration.

REMEMBER: Weak acid means partial ionisation in water; it does not mean dilute.

Alkanoic Acid Reaction Map



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

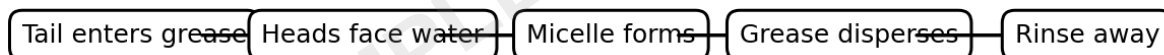
• 3.3 Detergents

INVESTIGATION: Compare soap action in soft and hard water using equal volumes and equal amounts of soap solution.

CAUTION: More foam does not automatically mean better cleaning or greater environmental safety.

REMEMBER: A detergent molecule has a hydrophobic region that interacts with grease and a hydrophilic region that interacts with water.

How Detergents Remove Grease



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

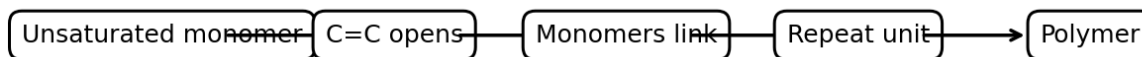
• 3.4 Polymers

THINK BIG: A good plastics plan connects durability to reduction, reuse, separation, recycling and controlled disposal.

REMEMBER: Monomer, repeat unit and polymer are related but are not the same thing.

HINT: To infer a simple addition-polymer monomer, identify the two backbone carbons and restore the C=C bond.

Addition Polymerisation



Peak Achievers visual summary - use the diagram to connect concepts before studying details.

FINAL REVISION & ASSESSMENT

- Mixed Concepts Tests 1-2
- National CBE Examination Models 1-5
- Marking Schemes
- Quick Reference
- About the Author

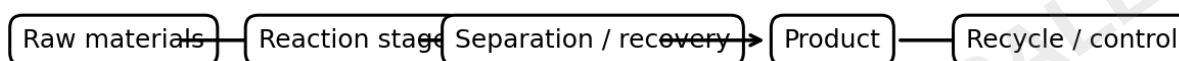
1.0 INORGANIC CHEMISTRY

1.1 Industrial Manufacture of Inorganic Chemicals

EXAM FOCUS: When explaining an industrial condition, use: CONDITION -> CHEMICAL REASON -> INDUSTRIAL ADVANTAGE.

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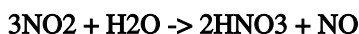
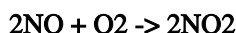
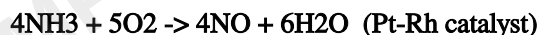
Industrial Manufacture - Process Thinking



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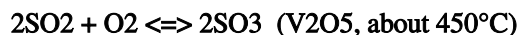
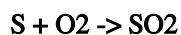
1.1.1 Ostwald Process - Manufacture of Nitric Acid

Ammonia is catalytically oxidised to nitric oxide, which is further oxidised and absorbed in water to form nitric acid.



Industrial control focuses on efficient oxidation, gas handling and limiting nitrogen-oxide emissions.

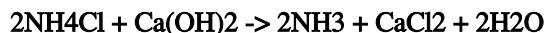
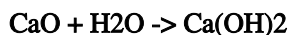
1.1.2 Contact Process - Manufacture of Sulphuric Acid



About 450°C is a compromise between a useful reaction rate and favourable equilibrium yield. The catalyst speeds attainment of equilibrium but does not shift equilibrium. SO_3 is absorbed in concentrated H_2SO_4 to avoid difficult acid mist.

1.1.3 Solvay Process - Manufacture of Sodium Carbonate

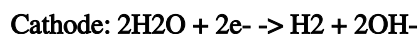
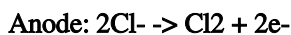




Ammonia and carbon dioxide are recycled where possible, reducing raw-material demand and waste.

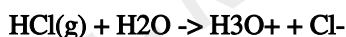
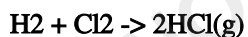
1.1.4 Chlor-Alkali Process

Concentrated brine is electrolysed in a membrane cell to produce chlorine, hydrogen and sodium hydroxide.



The membrane allows ionic conduction while helping keep products separated. Chlorine requires careful containment because it is toxic.

1.1.5 Industrial Manufacture of Hydrogen Chloride



Hydrogen chloride gas dissolves readily in water to form hydrochloric acid. Industrial handling must control corrosive fumes and prevent unsafe gas release.

1.1.6 Industrial Efficiency, Recycling and Environmental Responsibility

Industrial chemistry balances yield, reaction rate, energy use, raw-material recycling, safety and environmental impact. Efficient plants recover useful reactants, minimise waste and treat emissions or effluents before release.

NOTE: This section continues the completed Inorganic Chemistry sequence.

1.2 Carbon and Extraction of Metals

THINK BIG: Compare extraction from ore with recycling in terms of energy, resources, waste and environmental disturbance.

CAUTION: Limestone does not reduce iron ore. It provides CaO, which removes silica impurity as slag.

HINT: For extraction questions, first locate the metal relative to carbon in the reactivity series.

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